Quantum-Assisted Pathfinding for Warehouse Robots

Florida Qiskit Fallfest 2026

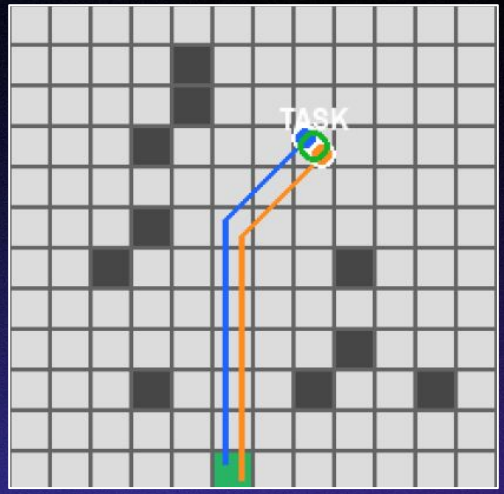
Team Quantum Robot Pathfinding

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The Problem

Autonomous robot in warehouses need to find efficient paths to their destinations while avoiding obstacles. Our goal is to find efficient efficient paths that can minimize travel distance and avoid obstacles while comparing classical and quantum pathfinding method.



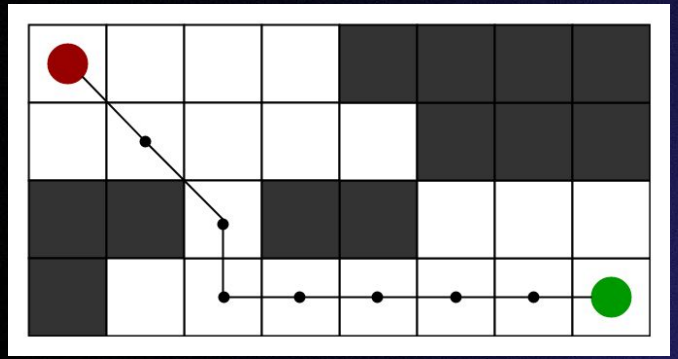


Image source: GeeksforGeeks, “A* Search Algorithm” (2026).

Our Approach

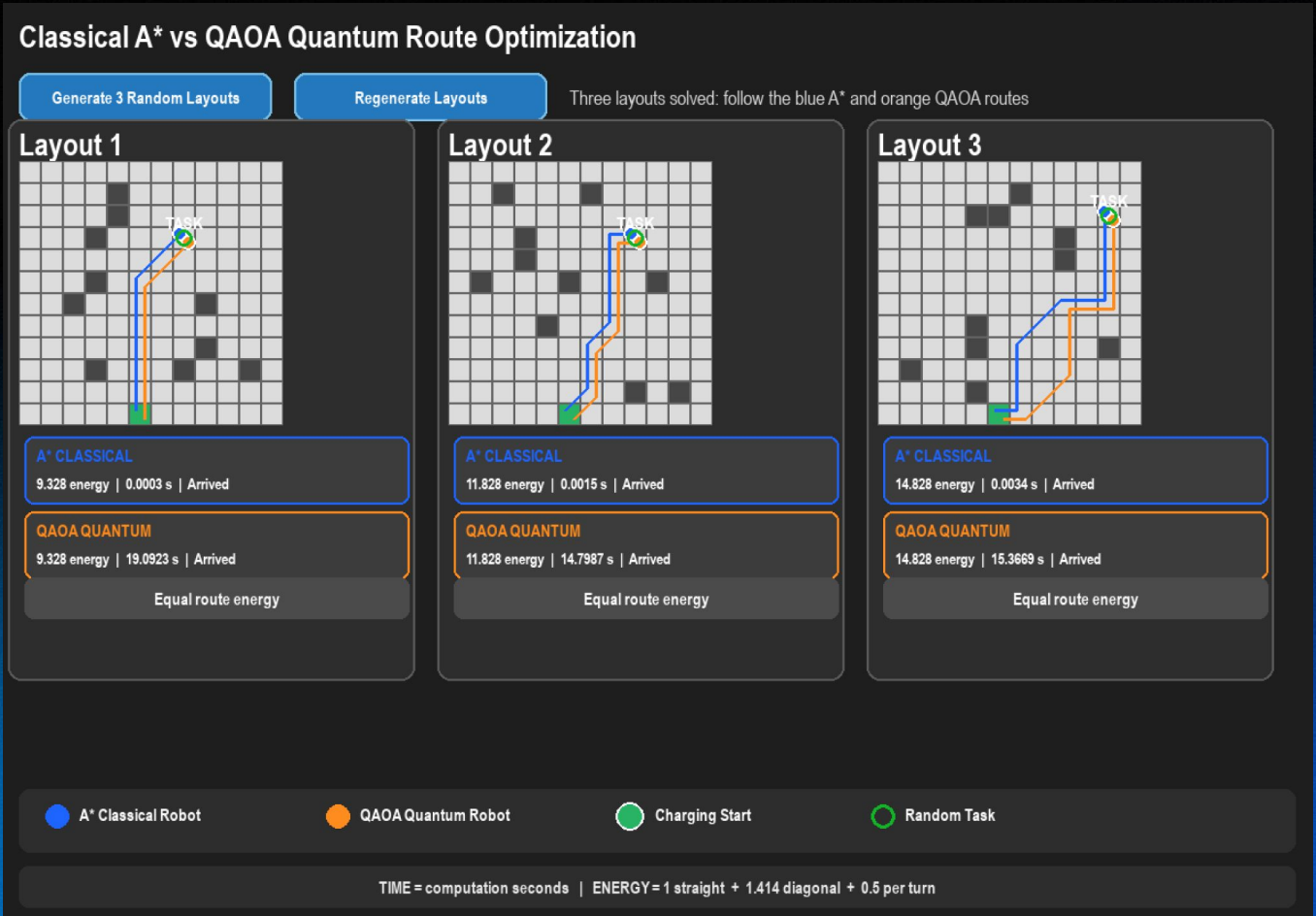
- Warehouse Simulation: Create a 12x12 grid with robots, obstacles, charging station as starting point, and destinations.
- Classical method: A* star pathfinding algorithm.
- Quantum Pathfinding: Qiskit and Quantum Approximate Optimization Algorithm (QAOA).
- Hardware Testing: Run a QAOA circuit on an IBM quantum computer using 1024 shots (measurements) and filter the results to identify valid routes.
- Comparison: Use 3 different random layouts to compare the computation time and energy.

Quantum Optimization

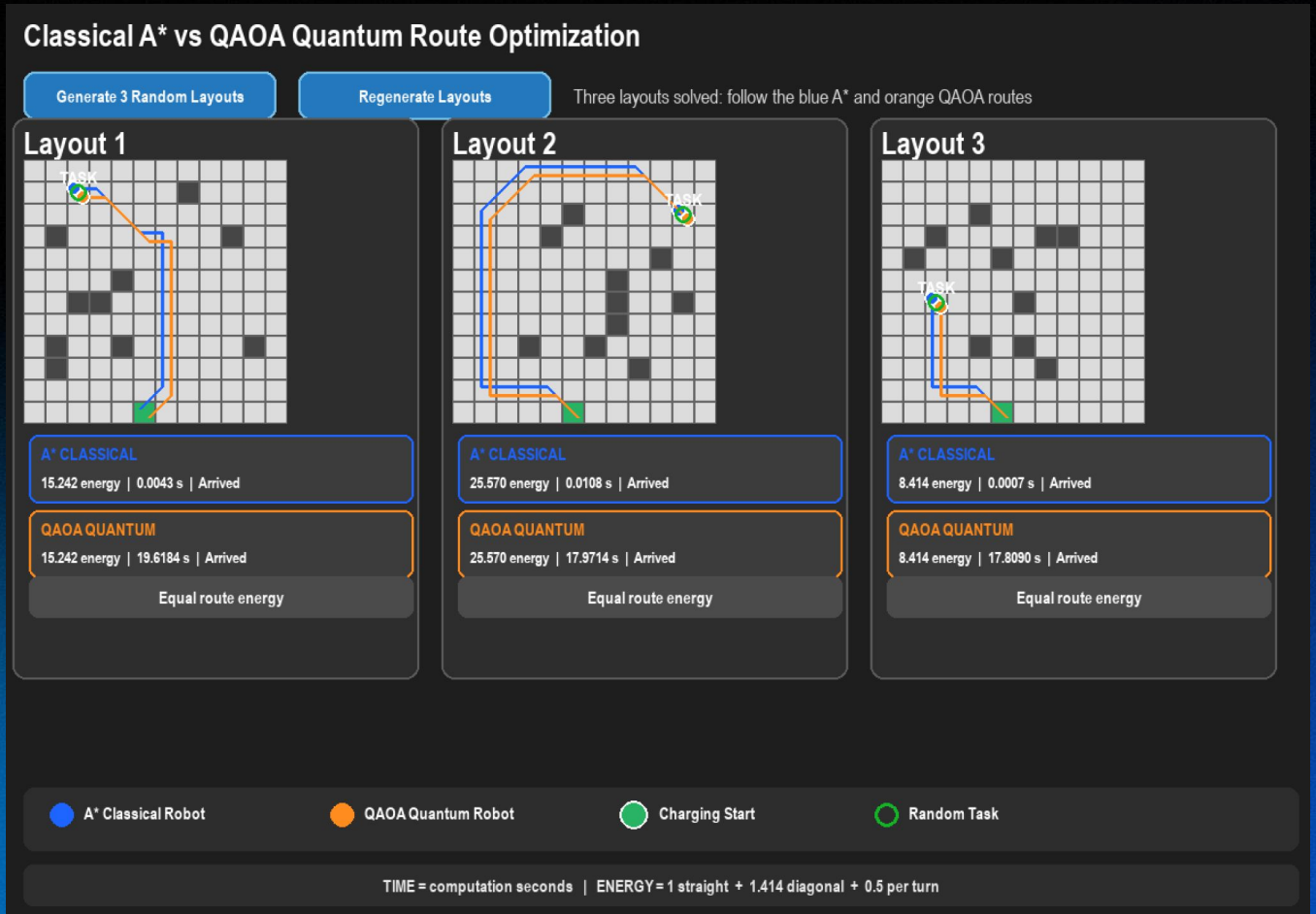
A* generates up to four possible routes with one baseline route while avoiding obstacles (5 in total). Each route has an energy cost based on movement and direction changes. In each run, QAOA starts with initial parameters, the optimizer adjusts the beta and gamma parameters, it searches for parameters that minimize the cost function, then it produces an optimized solution.

After the 5 runs after executed, the quantum hardware circuit is run 1024 times (shots), then picks the best route candidate. QAOA then explores different route selections, and the lowest-energy valid route is selected from the measurement results.

Classical vs Quantum Tests (1/2)



Classical vs Quantum Tests (2/2)



Classical

Run 1:

Layout 1:
- Energy: 9.328
- Computation time: 0.0003s

Layout 2:
- Energy: 11.828
- Computation time: 0.0015s

Layout 3:
- Energy: 14.828
- Computation time: 0.0034s

Run 2:

Layout 1:
- Energy: 15.242
- Computation time: 0.0043s

Layout 2:
- Energy: 25.570
- Computation time: 0.0108s

Layout 3:
- Energy: 8.414
- Computation time: 0.0007s

Quantum

Run 1:

Layout 1:
- Energy: 9.328
- Computation time: 19.0923s

Layout 2:
- Energy: 11.828
- Computation time: 14.7987s

Layout 3:
- Energy: 14.828
- Computation time: 15.3669s

Run 2:

Layout 1:
- Energy: 15.242
- Computation time: 19.6184s

Layout 2:
- Energy: 25.570
- Computation time: 17.9714s

Layout 3:
- Energy: 8.414
- Computation time: 17.8090s

Takeaways From Results:

Classical (all 6 layouts):
- Average energy: 14.2
- Average calculation time: 0.0035 s

Quantum (all 6 layouts):
- Average energy: 14.2 (same)
- Average calculation time: 16.41 s

- The quantum calculation took the same amount of energy (tiles on grid) to get to the goal.
- Quantum calculations were slower by a factor of 4,689 times bigger than the classical calculation times, most likely due to long connectivity time to the IBM quantum computer.
- The classical and quantum methods took slightly different paths to reach the goal, however they were the same amount of energy used when the methods were compared.

Limitations and Future Work

Limitations

- Long waiting times to access quantum hardware.
- Small number of test cases and limited problem size.

Future work

- Add more robots, larger warehouse grids, and more obstacles.
- Include battery management and charging optimization.
- Compare quantum optimization with traditional AI methods, such as reinforcement learning, to evaluate energy efficiency and computation time.

Conclusion

Classical calculations with the A* pathfinding are faster in this project due to a longer wait period to connect to the IBM quantum computer and calculations. However, the quantum calculations produced similar energy path usage with only slight variations with the directions that the paths took.